PRODUCT USER MANUAL

For Arctic Ocean - High Resolution Sea Ice Information L3 and L4

SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023
SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024
Issue: 1.1

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GLOSSARY AND ABBREVIATIONS

AI	Artificial Intelligence
AMSR2	Advanced Scanning Microwave Radiometer 2
ASIP	Automated Sea Ice Products
CMEMS	Copernicus Marine Environment Monitoring Service
CNN	Convolutional Neural Network
CF	Climate Forecast (convention for NetCDF)
DL	Deep Learning
DMI	Danish Meteorological Institute
(I)CDR	(Interim) Climate Data Record
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
FLZ	Floe Size
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
GLO	Global
OSI SAF	Ocean & Sea Ice Satellite Application Facility
L3	Level-3
L4	Level-4
MET	Norwegian Meteorological Institute
MDS	Marine Data Store
MFC	Monitoring and Forecasting Centre
MY	Multi Year

NERSC	Nansen Environmental Remote Sensing Centre
NetCDF	Network Common Data Form
NRT	Near Real Time
PUM	Product User Manual
QUID	Quality Information Document
RCM	Radarsat Constellation Mission
SAR	Synthetic Aperture Radar
SIC	Sea Ice Concentration
SOD	Stage-of-development
TAC	Thematic Assembly Center

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This product user manual (PUM) describes the *Arctic Ocean - High Resolution Sea Ice Information L3 and L4* products (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023 and SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_024) disseminated as part of the CMEMS Sea Ice Thematic Assembly Centre (SI TAC) since November 2024.
Figure 1 provide a schematic overview of the two products and four datasets covered by this document: A level-3 daily mosaic product of sea ice concentration, stage-of-development and floe size information with associated uncertainties, delivered as a near real time (NRT) dataset and a corresponding multiyear (MY, reprocessed) dataset extending the product back in time, covering the ten years of the Sentinel-1 mission from October 2014 to October 2024.
The collection also includes corresponding level-4 Arctic sea ice concentration product, where the level-3 datasets have been 'gap-filled' with OSI SAF microwave radiometer-based sea ice concentration datasets: the AMSR2 based OSI-458 climate data record (CDR) is used for the level-4 MY (reprocessed) dataset and the OSI-408 NRT sea ice concentration dataset is used for the level-4 NRT dataset. All four datasets are delivered in netcdf format with a polar stereographic projection; the level-3 product datasets on a 0.5 km spatial grid and the level-4 product datasets on a 1 km spatial grid.
The described products and datasets are results of a CMEMS service evolution project involving the Danish Meteorological Institute (DMI), Nansen Environmental Remote Sensing Centre (NERSC) and the Norwegian Meteorological Institute (MET). The service evolution goal was to further develop the DMI-ASIP AI-model used for producing the automated sea ice concentration products for Greenland waters (SEAICE_ARC_PHY_AUTO_L4_NRT_011_015, https://doi.org/10.48670/moi-00122, discontinued January 2025) and Antarctica (SEAICE_ANT_PHY_AUTO_L3_NRT_011_012, https://doi.org/10.48670/mds-00320) to provide high-resolution, detailed, consistent and standardized multivariate ice observations of the pan-Arctic with associated uncertainties for integration into forecast models.

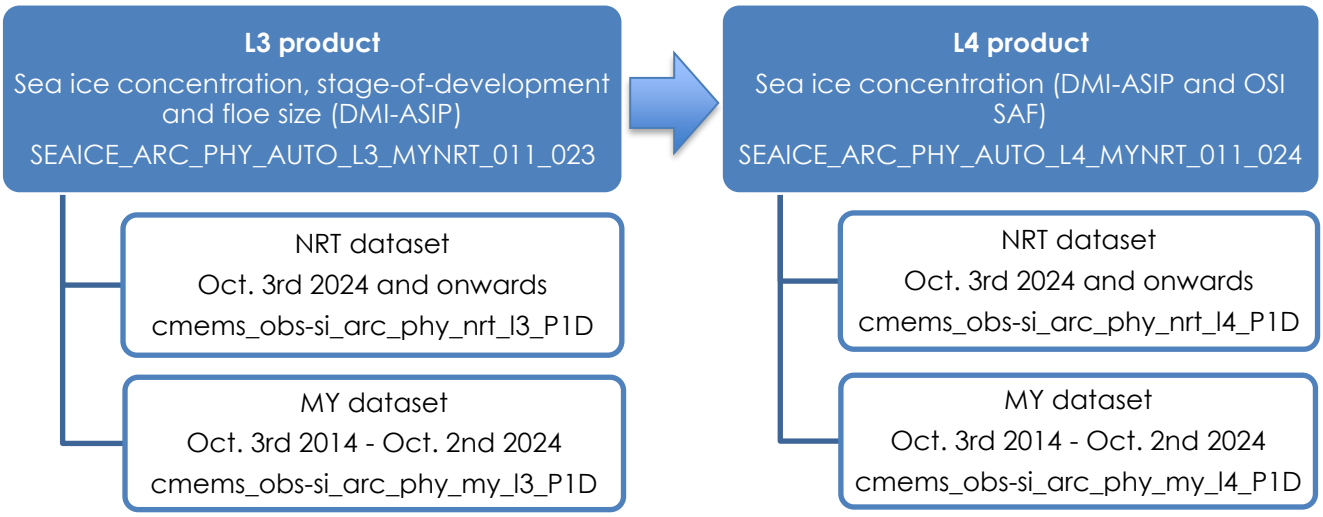

Figure 1 - Schematic overview of the two products and inherent four datasets covered by this Product User Manual. The L3 and L4 MYNRT (Multi-Year and Near-Real-Time) products, their inherent NRT and MY datasets and the time series they cover.

b) Data acknowledgement

The authors acknowledge the following data providers for the provision of datasets used as input to the products addressed in this PUM:

- ESA/Copernicus and the Copernicus Data Space Ecosystem (https://dataspace.copernicus.eu/) for the provision of Sentinel-1 SAR imagery.
- Alaska SAR facility (https://asf.alaska.edu/) for archive Sentinel-1 SAR imagery.
- The Earth Observation Data Management System (EODMS, https://www.eodms-sgdot.nrcan-rncan.gc.ca) for provision of Radarsat Constellation Mission SAR imagery.
- The Japanese Space Agency (JAXA) for AMSR2 microwave radiometer data (from https://gportal.jaxa.jp/gpr/index/index).
- The EUMETSAT Ocean and Sea Ice SAF for the OSI-458 Global Sea Ice Concentration v.3 Climate Data Record and the OSI-408-a Sea Ice Concentration (AMSR2) Level 3 Near Real Time product (https://osi-saf.eumetsat.int/products/sea-ice-products).
- The Norwegian Meteorological Institute for the OSI-458 ICDR product[1].

[1] The MET OSI-458 ICDR is an unofficial and non-operational dataset produced by MET also responsible for the EUMETSAT OSI SAF OSI-458 CDR. The ICDR has been produced using the same algorithm as the official CDR, with the purpose of serving as auxiliary input to a Copernicus Climate Change product. MET has kindly made the ICDR dataset available for DMI to use it as auxiliary input to the L4 MY dataset (cmems_obs-si_arc_phy_my_l4_P1D), extending the OSI-458 CDR.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Table 1 – Arctic Ocean High Resolution Sea Ice Information L3 general information.

Product Title	**Arctic Ocean - High Resolution Sea Ice Information L3**
Product Reference	SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023
Product Type	Multiyear and Near Real Time (MYNRT), Analysis
Description	Arctic L3 sea ice product providing concentration, stage-of-development and floe size information retrieved from Sentinel-1 SAR imagery and Radarsat Constellation Mission (RCM) and GCOM-W AMSR2 microwave radiometer data using the DMI-ASIP deep learning algorithm and delivered on a 0.5 km grid.
Model version	DMI-ASIP version 2.0
Input data source	Satellite observations: • Sentinel-1 (S-1) SAR EW and IW* imagery (*only for NRT dataset) • Radarsat Constellation Mission (RCM) SAR imagery (only NRT dataset) • GCOM-W AMSR2 microwave radiometer data
Dataset references	• cmems_obs-si_arc_phy_my_l3_P1D • cmems_obs-si_arc_phy_nrt_l3_P1D
NetCDF WMS titles	• DMI-ASIP sea ice information – Arctic MY L3 • DMI-ASIP sea ice information – Arctic NRT L3
Variables	• Sea ice concentration (sic) • Standard deviation (sic) • Sea ice stage-of-development category (sod) • Confidence (sod) • Sea ice floe size category (flz) • Confidence (flz) • Acquisition time (of S-1 imagery) • Status flag
Available time series	MY dataset : October 3rd 2014 to October 2nd 2024. NRT dataset : October 3rd 2024 and onwards.
Temporal resolution	24 hour daily mosaic (no time-averaging applied)
Update Frequency	MY dataset : Annual update, 1 year dataset extension. NRT dataset : Daily.
Target delivery time	Daily+1 at 12 pm (noon) UTC
Geographical coverage	Arctic: 180°W-180°E ; 40°S – 90°N
Horizontal resolution	0.5 km
Grid	Polar-stereographic
Vertical levels	Surface only
Format	NetCDF4

Table 2 – Arctic Ocean High Resolution Sea Ice Information L4 general information.

Product Title	**Arctic Ocean - High Resolution Sea Ice Information L4**
Product Reference	SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024
Product Type	Multiyear and Near Real Time (MYNRT), Analysis
Description	Arctic L4 sea ice concentration product based on the L3 sea ice concentration product (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) retrieved from Sentinel-1 and Radarsat Constellation Mission (RCM) SAR imagery and GCOM-W AMSR2 microwave radiometer data using a deep learning algorithm, gap-filled with OSI SAF EUMETSAT sea ice concentration products and delivered on a 1 km grid.
Model version	• DMI-ASIP version 2.0 • EUMETSAT OSI SAF OSI-458 CDR/ICDR* version 3.0 / OSI-408-a
Input data source	Satellite observations: • Sentinel-1 (S-1) EW and IW* imagery (*only for NRT dataset) • Radarsat Constellation Mission (RCM) SAR imagery (only NRT dataset) • GCOM-W AMSR2 microwave radiometer data
Dataset references	• cmems_obs-si_arc_phy_my_l4_P1D • cmems_obs-si_arc_phy_nrt_l4_P1D
NetCDF WMS titles	• DMI-ASIP sea ice information – Arctic MY L4 • DMI-ASIP sea ice information – Arctic NRT L4
Variables	• Sea ice concentration (SIC), DMI-ASIP / EUMETSAT OSI SAF • Uncertainty (SIC), DMI-ASIP / EUMETSAT OSI SAF • Status flag
Available time series	MY dataset : October 3rd 2014 to October 2nd 2024. NRT dataset : October 3rd 2024 and onwards.
Temporal resolution	24 hour daily mosaic (no time-averaging applied)
Update Frequency	MY dataset : Annual update, 1 year dataset extension. NRT dataset : Daily.
Target delivery time	Daily+1 at 12 pm (noon) UTC
Geographical coverage	Arctic: 180°W-180°E ; 40°S – 90°N
Horizontal resolution	1 km
Grid	Polar-stereographic
Number of vertical levels	Surface only
Format	NetCDF4

b) Production System Description

The *Arctic Ocean - High Resolution Sea Ice Information L3 and L4* products are produced operationally by the Production Unit at the Danish Meteorological Institute (DMI) and provided to CMEMS through the SI TAC Dissemination Unit at the Norwegian Meteorological Institute.
Two individual production systems have been established for the generation of the MY and NRT datasets, respectively, based on differences in input data and operational criteria. However, the two systems are alike in terms of processing steps from level-2 (L2) to level-3 (L3) and level-4 (L4). Figure 2 provide a schematic overview of the production system that is described step-wise in the following sections.

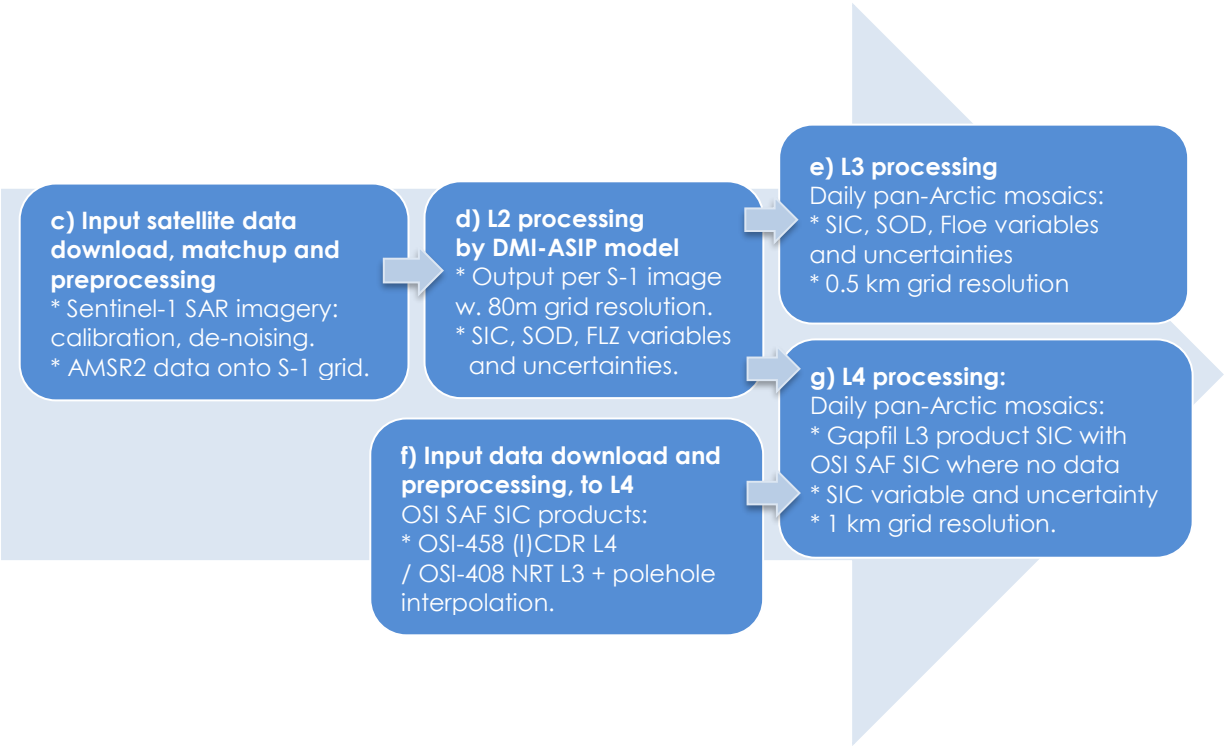

Figure 2 – Flow chart of the production system. The item letters in each box, starting from c), refer to the PUM section that describe the process: from download of (c) input satellite data and (f) auxiliary data products, to the production of (d) level-2 and hence the final (e) level-3 and (g) level-4 products for CMEMS.

c) Input Satellite Data

The DMI-ASIP model that is used for producing the sea ice products (and is described in the next section) takes SAR imagery as input, from the Copernicus Sentinel-1 mission and Radarsat Constellation Mission (RCM, since November 2025), combined with passive microwave data from the AMSR2 sensor on board the Japanese GCOM-W satellite.

SAR imagery from Sentinel-1 and Radarsat Constellation Mission

The DMI-ASIP model utilizes the Sentinel-1 level-1 (L1) Ground Range Detected Medium resolution (GRDM) scenes that are in Extra Wide (EW) and Interferometric Wide (IW) swath mode and with HH/HV polarizations (SAR signals transmitted in horizontal (H) polarisation only and then received in both horizontal (H) and vertical (V) polarisations). For more information about the Sentinel-1 sensor, see the SentiWiki home page[2].

The first Copernicus Sentinel satellite, Sentinel-1A, was launched on April 3rd 2014 and the first Sentinel-1A imagery available on the online data servers (Alaska Satellite Facility, as well as Copernicus Data Space Ecosystem) is for October 3rd 2014. The L3 and L4 MY datasets start on October 3rd 2014 and extend to October 2nd 2024, to complete a 10 year dataset time series. Sentinel-1B was launched April 25th 2016 and was operated in synergy with Sentinel-1A until it was lost in late December 2021. Thus, the dataset period from June 23rd 2016 to December 23rd 2021 where both Sentinel-1 satellites were operated has the best data coverage. Whereas the period from October 3rd 2014 to April 25th 2016 and from December 24th 2021 onwards, when only Sentinel-1A was/is operated, experiences a less good data coverage.

All archive Sentinel-1(A and B) imagery for use in the L3 and L4 MY datasets was downloaded effectively at the Alaska Satellite Facility, see Table 5. The near-real-time download of Sentinel-1 imagery for use in the L3 NRT dataset is downloaded at the Copernicus Data Space Ecosystem, see Table 3. A noticeable difference between the MY and NRT dataset's use of Sentinel-1 imagery, is that the MY dataset is based on EW mode imagery only, while the NRT dataset is based on both EW and IW mode imagery. IW mode imagery was not used as input to the ten-year MY dataset time series due to the additional processing resources that large amount of data would require.

The Sentinel-1 EW and IW imagery is preprocessed prior to the ingestion in the DMI-ASIP model; both image modes are resampled onto an 80 m grid. The 80 m grid resolution approximately matches the nominal spatial resolution of the Sentinel-1 imagery in Extra Wide (EW) swath acquisition mode. Data calibration and de-noising is applied using the noise vectors provided in the Sentinel-1 product metadata. Noise vectors were not provided for Sentinel-1 products prior to 2018, and thus, Sentinel-1 scenes from 2014-2018 contain a considerable amount of thermal noise compared to more recent acquisitions. However, the DMI-ASIP model performance is good even for the earliest S-1 data, where de-noising information is currently not available from ESA (DMI-ASIP model noise-handling strategies are explained in section d).

The DMI-ASIP model also utilizes, since November 2025, the Radarsat Constellation Mission (RCM) level-1 (L1) ScanSAR scenes that are in Medium Resolution (MR) and Low Noise (LN) beam modes and with HH/HV polarizations. For more information about the RCM sensor, see the RCM home page[3]. The RCM MR and LN imagery is preprocessed prior to the ingestion in the DMI-ASIP model; both image modes are resampled onto an 80 m grid. The 80 m grid resolution approximately matches the nominal spatial resolution of the RCM MR (50 m) and LN (100 m) acquisition modes. Data calibration and de-noising is applied using the same methodology as used for Sentinel-1 products.

AMSR2 microwave radiometer data

AMSR2 microwave radiometer Level-1b data, for use in both MY dataset and NRT dataset production has been/is downloaded at the JAXA Globe Portal System

[2] Copernicus Sentinel Missions Wiki home page: https://sentiwiki.copernicus.eu/web/sentinel-1 (last accessed 01-10-2024).

[3] Government of Canada Radarsat Constellation Mission home page: https://www.asc-csa.gc.ca/eng/satellites/radarsat/ (last accessed 20-11-2025).

(https://gportal.jaxa.jp/gpr/index/index). The DMI-ASIP model utilizes all available frequencies from the AMSR2 sensor (6.9GHz, 7.3 GHz, 10.7 GHz, 18.7 GHz, 23.8 GHz, 36.5 GHz, 89.0 GHz) in H/V polarization. The spatial resolutions of the AMSR2 channels range from a few kilometers to tens of kilometers depending on the frequency. The AMSR2 Level-1b swath products that intersect with the geographical extent bounded by the respective S1 and RCM image and that are acquired within maximum 7 hours of the S-1 image (limit set to avoid too much effect of potential sea ice drift) were resampled to a 2 km grid matching the geometry of the S1 and RCM image using a Gaussian weighted interpolation.

Table 3 – Input satellite data used in the production of the (L2 and hence the) Arctic Ocean - High Resolution Sea Ice Information L3 product; MY and NRT datasets, respectively.

Arctic Ocean - High Resolution Sea Ice Information L3 SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023	
Dataset	**Input satellite data**
cmems_obs-si_arc_phy_my_l3_P1D	Copernicus Sentinel-1 SAR imagery: • Level-1b, Extrawide (EW) swath mode, Ground range detected Medium Resolution (GRDM), polarizations: HH+HV. • Data period: Oct. 3rd 2014 to Oct. 2nd 2024 • Source: Alaska Satellite facility https://asf.alaska.edu/ JAXA GCOM-W AMSR2 data: • L1b, brightness temperatures • Period: Oct. 3rd 2014 to Oct. 2nd 2024 • Source: JAXA G-portal (ftp) https://gportal.jaxa.jp/gpr/index/index
cmems_obs-si_arc_phy_nrt_l3_P1D	Copernicus Sentinel-1 SAR imagery: • Data: Level-1b, Extrawide (EW) *and* interferometric wide (IW) swath mode, Ground range detected Medium Resolution (GRDM), polarizations: HH+HV. • Data period: Oct. 3rd 2024 and onwards • Source: Copernicus Data Space Ecosystem https://dataspace.copernicus.eu/ Radarsat Constellation Mission SAR imagery: • Data: Level-1b, Medium Resolution and Low Noise beam mode, ScanSAR, polarizations: HH+HV. • Data period: Nov. 4th 2025 and onwards • Source: Earth Observation Data Management System, www.eodms-sgdot.nrcan-rncan.gc.ca JAXA GCOM-W AMSR2 data: • Data: Level-1b, brightness temperatures • Period: Oct. 3rd 2024 and onwards • Source: JAXA G-portal (ftp) https://gportal.jaxa.jp/gpr/index/index

The NRT production system at DMI is set up to regularly check the data servers c.f. Table 3, for new available SAR satellite images covering the Arctic Ocean (dataset area). When a suitable SAR image is available the system checks the JAXA G-portal server for AMSR2 swath data the matches the S1 and RCM image in time and space. For more information about the input data match-up criteria, see [Wulf, 2024].
Throughout the day any new SAR imagery from Copernicus Data Space Ecosystem and the Earth Observation Data Management System (EODMS) and the corresponding latest AMSR2 swath data from the JAXA G-Portal are downloaded to DMI and automatically preprocessed individually.

d) Level-2 production by the DMI-ASIP deep learning model

The DMI-ASIP deep learning model is a Convolutional Neural Network (CNN) that performs data fusion of the SAR imagery with ~80 m pixel spacing with AMSR2 passive microwave radiometer (PMR) data, these sensors having complementary capabilities, to better resolve the ambiguities that can occur in SAR imagery e.g. at high wind speeds. Utilizing data from satellite sensors with different capabilities allow for ice products to be independent of daylight, more robust to different weather conditions and seasonal variation and result in a significant increase in product temporal frequency and geographical coverage compared to the existing ice chart products.
The automated interpretation of sea ice and water signatures in the two satellite data sources is done using the DMI-ASIP model that has been trained with regional ice charts produced at the DMI Greenland Ice Service and the Canadian Ice Service (CIS). The ice charts contain information about sea ice concentration, stage-of-development and floe size classes, following the World Meteorological Organization's (WMO) sea ice nomenclature. The training dataset was generated by spatio-temporally matching regional ice charts from DMI and CIS with Sentinel-1 imagery. A total of 5382 Sentinel-1 images and corresponding ice charts, spanning the years 2018 up to 2021, were used in the training of the DMI-ASIP model version 2.0. While the CNN has been trained on ice charts covering the Greenland and Canadian part of the Arctic the algorithm has been modified to run for the pan-Arctic region.
A first version of the DMI-ASIP model (v 1.0) is in use at DMI for producing the CMEMS *Antarctic Ocean High Resolution Sea Ice Information* product (SEAICE_ANT_PHY_AUTO_L3_NRT_011_012, https://doi.org/10.48670/mds-00320). The DMI-ASIP model version 1.0 has been trained with the ASIP/AI4Arctic Sea Ice Training Dataset (ASIDv2, https://doi.org/10.11583/DTU.13011134.v3) and the model has been generalized for the Southern Ocean. See [011_012 PUM, 2022] for details. The product contains information about sea ice concentration and associate uncertainties, only. A coming evolution of the *Antarctic Ocean - High Resolution Sea Ice Information* product is to update it to the DMI-ASIP model v2.0 for the sea ice concentration variable and make the dataset consistent with the *Arctic Ocean - High Resolution Sea Ice Information L3* product.
Previous versions of the training dataset are published as the ASIP/AI4Arctic Sea Ice Training Dataset (ASIDv2, https://doi.org/10.11583/DTU.13011134.v3) and the AI4Arctic Sea Ice Challenge Dataset (https://doi.org/10.11583/DTU.c.6244065.v2).
A full description of the training and development of the DMI-ASIP model version 2.0 for prediction of pan-Arctic sea ice concentration can be found in [Wulf, 2024(1)]. The model architecture is a modified multi-headed UNet for simultaneous retrieval of the three sea ice parameters (that are given from the ice charts); sea ice concentration

(SIC), stage-of-development (SOD) and floe size (FLZ). The methodology behind the multi-parameter retrieval, as well as a description of the training procedure that allows the model to handle noisy Sentinel-1 imagery (from before 2018, where ESA noise vectors are not yet provided), can be found in [Wulf, 2024(2)].

Sea ice parameters

From more than six years of focused development efforts, the DMI-ASIP model now uses a robust methodology for sea ice concentration retrieval. The sea ice concentration parameter predictions are output from 0 to 100%. See the dataset variable description in Table 5 and Figure 3 for an example of a level-2 sea ice concentration output.
The SOD categories, however, can be somewhat difficult to discriminate accurately based purely on observations (SAR/AMSR2) due to some categories having similar microwave signatures as well as the ambiguities stemming from using ice charts as label data (ice charts uses partial concentrations as labels i.e. each pixel containing multiple SOD categories). When ice analysts assign partial concentrations of SODs to the polygons in an ice chart, their analysis is primarily based on the available satellite observations, but they also rely on their extensive experience and knowledge about the 'typical' (or climatological) sea ice conditions in specific locations at specific times of the year. To emulate this, a bi-weekly 'sea ice climatology' has been derived from 15 years (2007-2022) of pan-Arctic ice charts produced by the US National Ice Center (NIC), for the DMI-ASIP model to use as its 'experience'. This resulted in an improved predictive performance for SOD.

The sea ice SOD is retrieved for 5 different classes:

- New and Young Ice (<30 cm)
- Thin First Year Ice (30 cm - 70 cm)
- Thick First Year Ice (>70 cm)
- Multi-Year Ice
- Glacier Ice

See the dataset variable description in Table 5 and the stage-of-development map plot in Figure 5.
The challenges mentioned for SOD above also applies to the discrimination and prediction of the Floe size (FLZ) categories. However, no auxiliary datasets where available for producing a suitable FLZ 'climatology'.

The sea ice FLZ is retrieved for 4 different classes:

- Small Floes (<100 m)
- Medium Floes (100 m - 500 m)
- Big Floes (500 m - 2 km)
- Vast and Giant Floes (>2 km)

See dataset variable description in Table 5 and the floe size map plot in Figure 6.

Sea ice parameter filtering

The 15-year sea ice 'climatology' based on pan-Arctic NIC ice charts is used also in the post-processing of the level-2 products for a 'maximum sea ice extent climatology screening': If SIC > 0% occur in a grid cell more than 50 km away from the sea ice climatology maximum sea ice extent (derived for every week of the year), the grid

cell SIC is set to 0%. The affected grid cells are marked with the status flag 'maximum sea ice extent climatology screening'. See section i) for more information.
Likewise, the sea ice climatology is applied as a filter in the SOD variable: If either of the SOD classes Multi-Year Ice or Thick First Year Ice (>70 cm) occurs in a grid cell where that SOD class is not represented in the sea ice climatology, that grid cell SOD class is changed to the next most likely SOD class (predicted by the DMI-ASIP model). (If the same is the case for this SOD class, it is changed to the next most probable class, etc., until the obtained SOD is represented by the climatology). The affected grid cells are marked with a status flag 'SOD changed according to SOD-climatology'. See section i) for more information.

For low sea ice concentrations, the SAR image textures do not allow for a robust retrieval of stage-of-development and floe size information. Thus, SOD and FLZ information are only provided in grid cells where the SIC variable is >= 15%. The SOD and FLZ dataset variables have a no-data fill value in grid cells where SIC < 15%.

Uncertainty estimates

For sea ice retrievals provided by deep learning-based retrieval models to be used in safety-critical applications, such as in the context of operational sea ice charting and sea ice information dissemination, or for data assimilation into numerical ocean-sea-ice forecast models, it is a prerequisite that the retrievals are accompanied by meaningful uncertainties. Along with the sea ice parameter (SIC, SOD, FLZ) class predictions, the deep learning model provides uncertainties/confidence scores. Various calibration strategies have been used to calibrate the DMI-ASIP model and to derive meaningful uncertainties from the confidence scores (see [Wulf, 2024(1)]) for the three sea ice parameters.
The L2 stage of the production system takes in the SAR image and matching AMSR2 data on the SAR image geometry 80m spatial grid and runs it through the DMI-ASIP model, that outputs - on the same 80m grid - the three sea ice parameters – SIC, SOD, FLZ, and their associated uncertainties. See Figure 3 for an example of a level-2 sea ice concentration output.

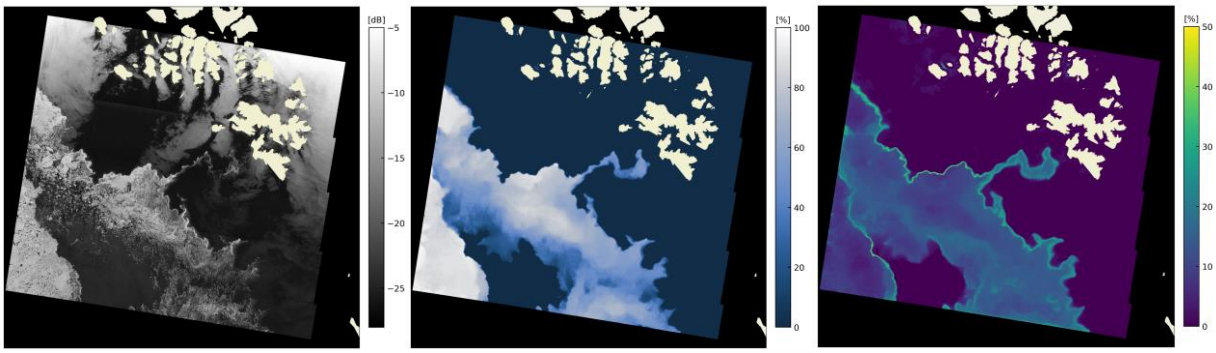

Figure 3 – Example of a DMI-ASIP model level-2 (L2) sea ice concentration output (on 80m spatial grid)[middle plot] and the uncertainty (standard deviation) on the sea ice concentration [right plot], based on a Sentinel-1 image [left plot] from September 9th 2021.

e) Level-3 production

After midnight (UTC), when the last SAR imagery acquired during the day have been processed into L2 products, the individual L2 products on 80 m grid are merged into a daily L3 mosaic dataset (newest information on top), with L2 product variables resampled onto a 0.5 km polar-stereographic grid and including information of satellite acquisition time given for each grid cell. No temporal interpolation (time averaging) nor spatial interpolation has been applied across the L2 products in the processing to L3 products (thus, transitions in sea ice parameter values might occur across some L2 swaths, with large difference in the acquisition time or high ice drift).
See example map plots of the L3 products three sea ice parameters, sea ice concentration, stage-of-development and floe size, and their associated uncertainties in Figure 4, Figure 5 and Figure 6, respectively.

f) EUMETSAT OSI SAF sea ice concentration products for Level-4 product

For the purpose of producing the Arctic Ocean - High Resolution Sea Ice Information L4 product from the DMI-ASIP L3 product, different sea ice concentration products provided by EUMETSAT Ocean & Sea Ice Satellite Application Facility (OSI SAF) have been used as input for gap-filling of the L3 product. The L4 product contains sea ice concentration information and associated uncertainty only.
The OSI SAF OSI-458 Climate Data Record (CDR) and Interim CDR (ICDR)[3], is used as input to the L4 MY dataset (cmems_obs-si_arc_phy_my_l4_P1D), whereas the OSI-408-a L3 NRT product is used for the L4 NRT dataset (cmems_obs-si_arc_phy_nrt_l4_P1D). An overview of the products used is provided in **Erreur ! Source du renvoi introuvable.**.

g) Level-4 production

After midnight (UTC), when all the SAR images acquired during the day have been processed and are available as L2 products, the L3 and L4 products are generated in parallel, and both take the individual L2 products as input.
Where the L3 product resamples the 80m grid L2 products to a 0.5 km polar-stereographic grid, the L4 product resamples the individual L2 products from one day onto a 1.0 km polar stereographic grid. Only the sea ice concentration and the associated uncertainty in the L2 products are used in the L4 production. See L4 product general description in Table 2.
The OSI SAF sea ice concentration product is resampled onto the same 1.0 km polar stereographic grid and OSI SAF sea ice concentrations are used to fill the gaps where the DMI-ASIP 1.0 km product does not provide sea ice concentration information.
Whereas the OSI-458 CDR/ICDR products used for the L4 MY dataset are L4 and gap-free, the OSI-408 L3 NRT that is used for the L4 NRT dataset has no data in the pole-hole. See [OSI SAF NRT PUM, 2023]. Thus, an additional processing step is added to the L4 NRT production where the OSI-408 dataset pole-hole grid cells are interpolated with sea ice concentration information, using a bilinear method. For the corresponding pole-hole in the uncertainty variable, all grid cells are set to 100% uncertainty. In case of missing data in the OSI-408 L3 NRT dataset due to missing AMSR2 swath data, the data gap is filled using the L4 NRT dataset from the day before (and the time variable for the affected grid cells are set accordingly to the date of the day before).

Table 4 – Input datasets used in the production of the Arctic Ocean - High Resolution Sea Ice Information L4 product; MY and NRT datasets, respectively.

Arctic Ocean - High Resolution Sea Ice Information L4 SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024	
Dataset	**Input datasets**
cmems_obs-si_arc_phy_my_l4_P1D	Arctic Ocean - High Resolution Sea Ice Information L3, SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023: • cmems_obs-si_arc_phy_my_l3_P1D EUMETSAT OSI SAF OSI-458 CDR version 3.0 • L4 sea ice concentration and uncertainties • Period: Oct. 3rd 2014 to Dec. 31st 2020 • Source: EUMETSAT OSI SAF https://osi-saf.eumetsat.int/products/osi-458 MET OSI-458 ICDR version 3.0 • L4 sea ice concentration and uncertainty • Period: Jan. 1st 2021 to Oct. 2nd 2024 • Source: Norwegian Meteorological Institute (MET), unofficial release[4]
cmems_obs-si_arc_phy_nrt_l4_P1D	Arctic Ocean - High Resolution Sea Ice Information L3, SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023: • cmems_obs-si_arc_phy_nrt_l3_P1D EUMETSAT OSI SAF OSI-408-a NRT • L3 sea ice concentration and uncertainties • Period: Oct. 3rd 2024 onwards • Source: EUMETSAT OSI SAF https://osi-saf.eumetsat.int/products/osi-408-a

A common land mask has been produced by combining the high-resolution DMI-ASIP land mask and the coarse resolution landmasks of both OSI-458 CDR/ICDR and the OSI-408. The coarser OSI SAF landmasks cover the DMI-ASIP landmask in most areas except where there are small islands and archipelagos that were not resolved in the OSI SAF landmasks. Few islands are not represented in the fine resolution DMI-ASIP land mask and thus might result in the DMI-ASIP model predicting sea ice in those grid cells. The common landmask is consistent for the L4 MY dataset, however it may change in the L4 NRT dataset in summer months, due to seasonal changes with the OSI-408 landmask. For more information, see [OSI SAF NRT PUM, 2023].
Relevant status flags in the OSI SAF products are imported to the DMI-ASIP L4 datasets. The list of status flags in the L4 datasets is given in Table 8.
The daily L4 mosaic product is stored as a NetCDF file and pushed to the CMEMS dissemination Unit.

[4] The MET OSI-458 ICDR is an unofficial and non-operational dataset produced by MET also responsible for the EUMETSAT OSI SAF OSI-458 CDR. The ICDR has been produced using the same algorithm as the official CDR, with the purpose of serving as auxiliary input to a Copernicus Climate Change product. MET has kindly made the ICDR dataset available for DMI to use it as auxiliary input to the cmems_obs-si_arc_phy_my_l4_P1D dataset.

h) Level-3 and Level-4 dataset details

The primary variables of the L3 datasets are the sea ice parameters – sea ice concentration (SIC), stage-of-development (SOD) and floe size (FLZ) - and their associated uncertainty information (standard deviation, confidence). Primary variables of the L4 datasets are the sea ice concentration (SIC) and associated uncertainty information (standard deviation).
In addition, both the L3 and L4 datasets contain auxiliary variables that describe the common grid: a regular Polar stereographic coordinate reference system (crs), spanned by x- (xc) and y-coordinates (yc) that define the grid for the latitudes (lat) and longitudes (lon).

Table 5 – Arctic Ocean High Resolution Sea Ice Information L3 dataset detailed information: variable names and description. For a full overview of the dataset netcdf file header information see Annex ***Erreur ! Source du renvoi introuvable.****.*

Arctic Ocean - High Resolution Sea Ice Information L3 SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023 cmems_obs-si_arc_phy_my_l3_P1D and cmems_obs-si_arc_phy_nrt_l3_P1D	
Dataset Variable	**Description**
sic	• long_name: sea ice concentration • standard_name: sea_ice_area_fraction • units: % • description: DMI-ASIP Sea ice concentration (sea ice area fraction) in percent (%) [0-100]. • FillValue: -1
standard_ deviation_ sic	• long_name = standard deviation for the predicted sea ice concentration • units: % • FillValue: -1
sod	• long_name = sea ice stage of development category • units = 1 • *description = DMI-ASIP Stage of development category IDs:* *0: New and Young Ice (<30 cm)* *1: Thin First Year Ice (30 cm - 70 cm)* *2: Thick First Year Ice (>70 cm)* *3: Multi-Year Ice* *4: Glacier Ice* • FillValue = -1
confidence_s od	• long_name = confidence level for the predicted stage of development • units = % • FillValue = -1
flz	• long_name = sea ice floe size category • units = 1 • *description = Floe size category IDs:* *0: Small Floes (<100 m)* *1: Medium Floes (100 m - 500 m)* *2: Big Floes (500 m - 2 km)*

	3: Vast and Giant Floes (>2 km) • FillValue = -1
confidence_flz	• long_name = confidence level for the predicted floe size category • units = % • source = DMI-ASIP • FillValue = -1
acq_time	• long_name = acquisition time (seconds since 1970-01-01 00:00:00) • units = s • FillValue = 0.0f; // float
Status flag	• standard_name = status_flag • units = 1 • flag_description: *See Section 0 for details.*
time	• standard_name = time • units = days since 1970-01-01 00:00:00 • calendar = standard
crs	• long_name = coordinate reference system • grid_mapping_name = polar_stereographic • proj4_string = +proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +x_0=0 +y_0=0 +a=6378273 +b=6356889.449 +units=m +no_defs
xc	• long_name = x-coordinate in cartesian system • standard_name = projection_x_coordinate • axis = X • units = m
yc	• long_name = y-coordinate in cartesian system • standard_name = projection_y_coordinate • axis = Y • units = m
lat	• long_name = latitude coordinate • standard_name = latitude • least_significant_digit = 3 • units = degrees_north
lon	• long_name = longitude coordinate • standard_name = longitude • least_significant_digit = 3 • units = degrees_east

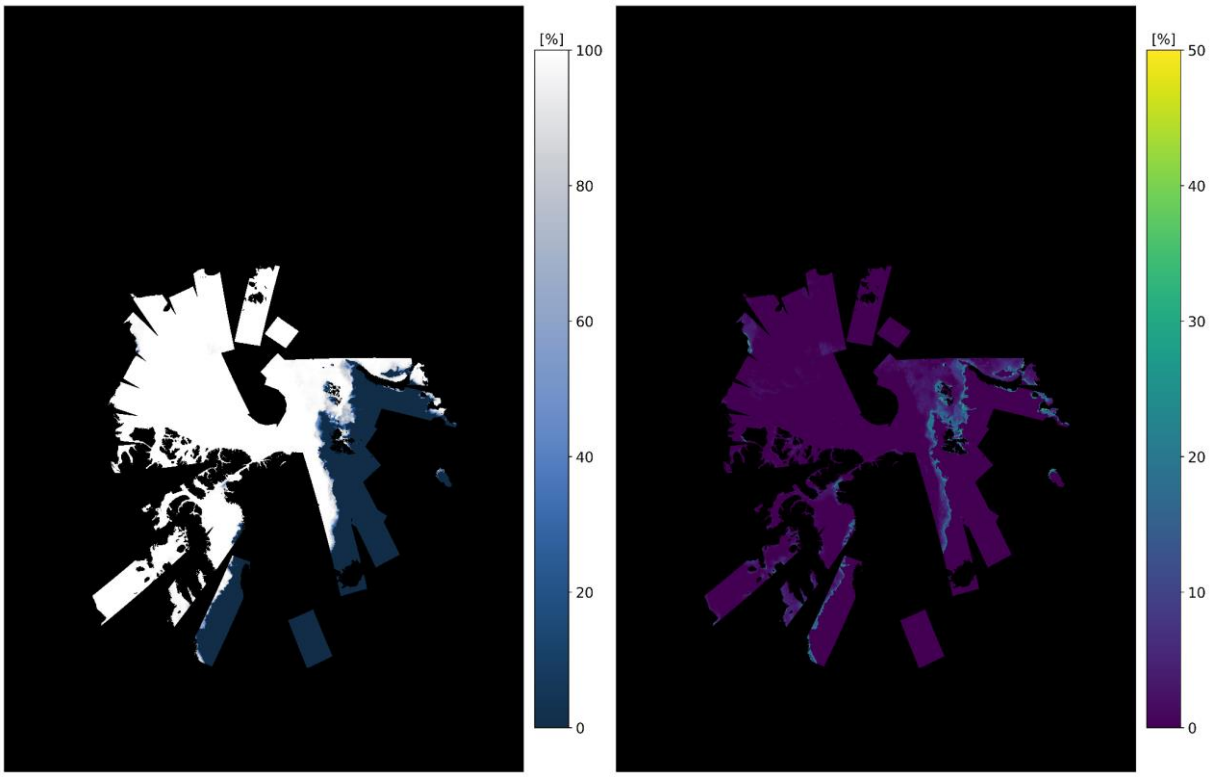

Figure 4 - Map plot of the Arctic Ocean High Resolution Sea Ice Information L3 product (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) daily dataset file for January 1st 2021. [left:] the sea ice concentration (SIC) variable and [right:] the standard deviation uncertainty on the sea ice concentration.

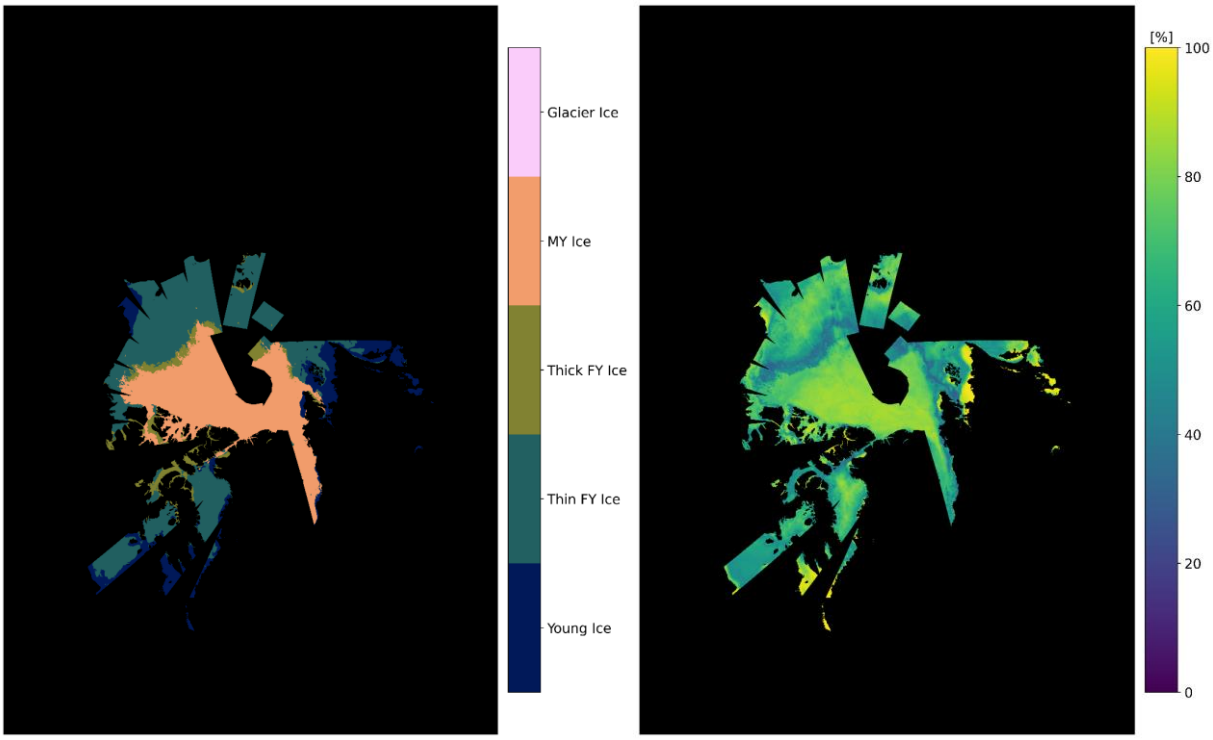

Figure 5 - Map plot of the Arctic Ocean High Resolution Sea Ice Information L3 product (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) daily dataset file for January 1st 2021. [left:] the sea ice stage-of-development (sod) variable and [right:] the confidence in the SOD categories.

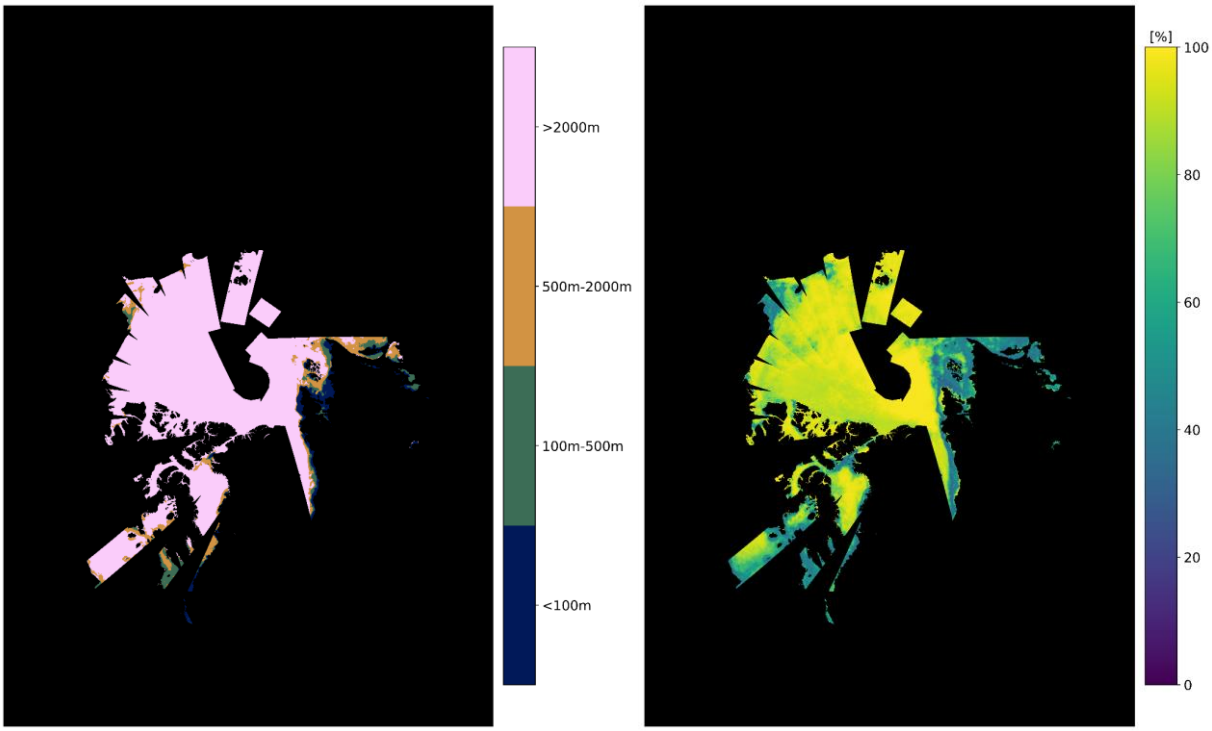

Figure 6 - Map plot of the Arctic Ocean High Resolution Sea Ice Information L3 product (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) daily dataset file for Jan. 1st 2021. [left:] the sea ice floe size (FLZ) variable and [right:] the confidence in the FLZ categories.

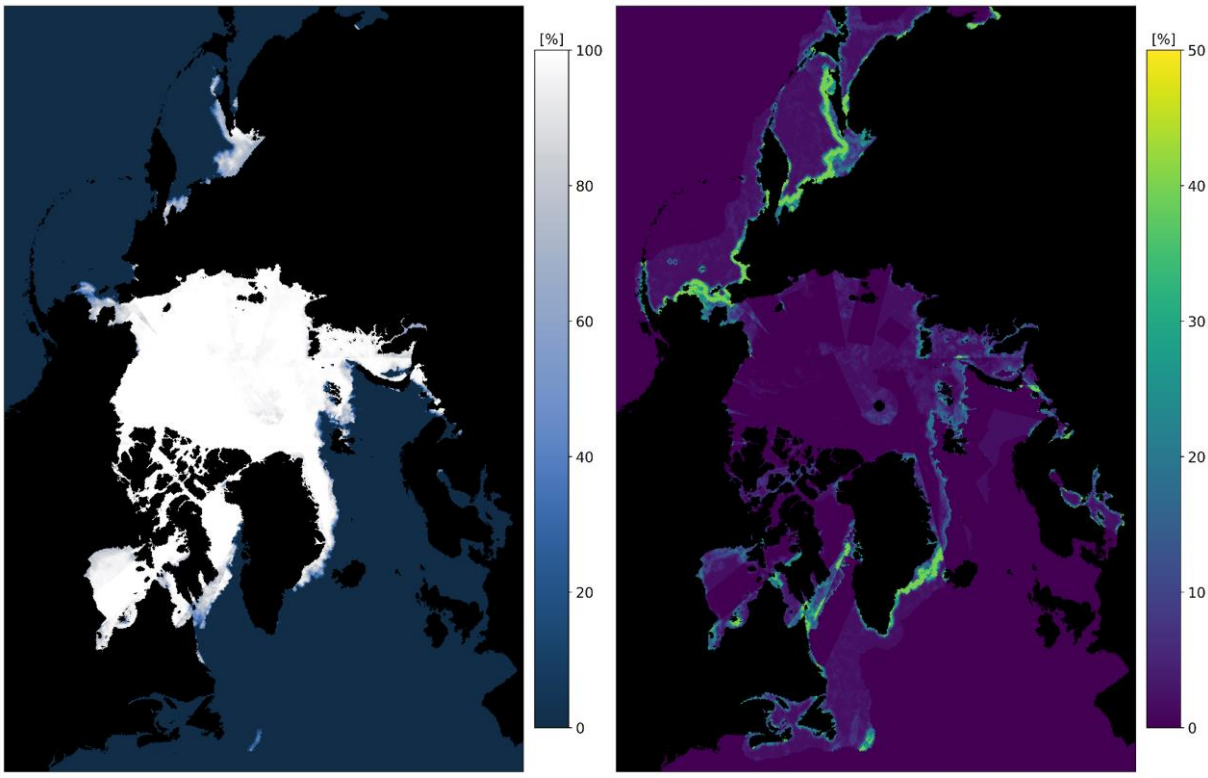

Figure 7 - Map plot of the Arctic Ocean High Resolution Sea Ice Information L4 product (SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024) daily MY dataset file for January 1st 2021. [left:] the sea ice concentration (SIC) variable and [right:] the standard deviation uncertainty on the sea ice concentration.

Table 6 – Arctic Ocean High Resolution Sea Ice Information L4 dataset detailed information: variable names and description. For a full overview of the dataset netcdf file header information see Annex ***Erreur ! Source du renvoi introuvable.****.*

Arctic Ocean - High Resolution Sea Ice Information L4 SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024 cmems_obs-si_arc_phy_my_l4_P1D and cmems_obs-si_arc_phy_nrt_l4_P1D	
Dataset Variable	**Description**
sic	• long_name = sea ice concentration • standard_name = sea_ice_area_fraction • units = % • description: DMI-ASIP Sea ice concentration (sea ice area fraction) in percent (%) [0-100]. • FillValue = -1
uncertainty_sic	• long_name = standard deviation for the predicted sea ice concentration • units = "%"; • FillValue = -1
Status flag	• standard_name = status_flag • units = 1 • flag_description: *See Section 0 for details.*
time	• standard_name = time • units = days since 1970-01-01 00:00:00 • calendar = standard
crs	• long_name = coordinate reference system • grid_mapping_name = polar_stereographic • proj4_string = +proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +x_0=0 +y_0=0 +a=6378273 +b=6356889.449 +units=m +no_defs
xc	• long_name = x-coordinate in cartesian system • standard_name = projection_x_coordinate • axis = X • units = m
yc	• long_name = y-coordinate in cartesian system • standard_name = projection_y_coordinate • axis = Y • units = m
lat	• long_name = latitude coordinate • standard_name = latitude • least_significant_digit = 3 • units = degrees_north
lon	• long_name = longitude coordinate • standard_name = longitude • least_significant_digit = 3 • units = degrees_east

i) Status flags

Status flags are provided as a separate variable, coded as a "bit mask" that contains information about the character of the data in each grid cell e.g. masks applied or processing steps that have influenced the sea ice parameter value.
The status_flag "bit mask" enables that more than one bit can be set at the same time, e.g: '00110' which means that bit 2 and bit 3 are on. When searching for a specific screening test it is recommended to search for the corresponding bit and not for the corresponding status_flag integer value. For instance, if searching for status_flag = 2, which means that bit 2 is on ('00010'), the mixed cases such as status_flag = 6 (that means '00110' – where both bit 2 and bit 3 is on) will be missed.
The status flag values are different for the L3 and L4 products, due to the products' different input data. Status_flag information that only applies to the grid cells with DMI-ASIP data is marked with prefix [DMI-ASIP]. Status_flag information imported with the OSI SAF product and that only apply to the grid cells with OSI SAF data in it, has been marked with [OSI SAF]. Due to the use of two different OSI SAF SIC products for the L4 product MY and NRT datasets (the OSI-458 (I)CDR and the OSI-408, respectively) the status_flag information in the L4 products' MY and NRT datasets bit no. 4, 5, 7 and 10 are different, but related. Common status_flag information, such as the land mask, does not have a prefix. An overview of the status flags for the L3 and L4 products are described in Table 7 and Table 8, respectively.

Table 7 – Definition of status flag bits for the Arctic Ocean - High Resolution Sea Ice Information L3 (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) product.

Arctic Ocean - High Resolution Sea Ice Information L3 SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023 cmems_obs-si_arc_phy_my_l3_P1D and cmems_obs-si_arc_phy_nrt_l3_P1D		
Bit no. (nbit)	**Value**	**Flag description**
0 (0)	0	No bits set: nominal value of the sea ice parameters (sic, sod, flz) retrieved by DMI-ASIP
1 (1)	1	Land mask
2 (10)	2	[NRT] Based on RCM SAR imagery (else, based on Sentinel-1 SAR imagery).
3 (100)	4	*Not in use*
4 (1000)	8	*Not in use*
5 (10000)	16	*Not in use*
6 (100000)	32	*Not in use*
7 (1000000)	64	*Not in use*
8 (10000000)	128	*Not in use*
9 (100000000)	256	*Not in use*
10 (1000000000)	512	SOD changed according to SOD-climatology
11 (10000000000)	1024	Maximum sea ice extent climatology screening applied
12 to 16	…	*Not in use*

Table 8 – Definition of status flag bits for the Arctic Ocean - High Resolution Sea Ice Information L4 product (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_024). Status_flag information that only apply to the grid cells with DMI-ASIP data is marked with prefix [DMI-ASIP]. Status_flag information imported with the OSI SAF product and that only apply to the grid cells with OSI SAF data in it, has been marked with [OSI SAF]. Due to the use of two different OSI SAF SIC products for the MY and NRT datasets (the OSI-458 (I)CDR and the OSI-408, respectively) the status_flag information in bit no. 4, 5, 7 and 10 are different, but related, in the MY and NRT datasets. Common status_flag information, such as the land mask, does not have a prefix.

Arctic Ocean - High Resolution Sea Ice Information L4 SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024 cmems_obs-si_arc_phy_my_l4_P1D and cmems_obs-si_arc_phy_nrt_l4_P1D		
Bit no. (nbit)	**Value**	**Flag description**
0 (0)	0	No bits set: nominal value of the sea ice concentration (SIC) retrieved by DMI-ASIP.
1 (1)	1	Land mask
2 (10)	2	[OSI SAF] Lake mask
3 (100)	4	[OSI SAF] SIC is set to zero by the open water filter.
4 (1000)	8	MY: [OSI SAF] SIC value is changed for correcting land spill-over effects. NRT: [OSI SAF] Near coast grid points.
5 (10000)	16	MY: [OSI SAF] Handle with caution, the 2m air temperature is high at this position, and this might be false ice. NRT: [OSI SAF] NWP skin temperature screening applied
6 (100000)	32	[OSI SAF] Value is the result of spatial interpolation.
7 (1000000)	64	MY: [OSI SAF] Value is the result of temporal interpolation. NRT: [OSI SAF] Value is from the previous day
8 (10000000)	128	[OSI SAF] SIC is set to zero since position is outside maximum sea ice climatology.
9 (100000000)	256	Gapfilled with OSI SAF level-3 sea ice concentration.
10 (1000000000)	512	MY: *Not in use* NRT: [DMI-ASIP] Value is from the previous day.
11 (10000000000)	1024	[DMI-ASIP] Maximum sea ice extent climatology screening applied.
12 to 16	…	*Not in use*

j) Dataset files not processed

A few daily dataset files have not been processed due to no available input data for the specific date, e.g. lack of input Sentinel-1 imagery for the L3 product (that will result in no processing of the L4 product for the same date) or unavailability of an OSI SAF sea ice concentration product used as input to the L4 product. Those dates where dataset files have not been processed are listed in Table 9 and Table 10. The tables will be updated accordingly in the event that a NRT dataset cannot be processed in the future.

Table 9 – Dataset files missing for the Arctic Ocean - High Resolution Sea Ice Information L3 (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023) product.

Arctic Ocean - High Resolution Sea Ice Information L3 SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023 cmems_obs-si_arc_phy_my_l3_P1D and cmems_obs-si_arc_phy_nrt_l3_P1D	
Date of dataset file [YYYYMMDD]	**Note**
20141015 20141016 20141119	Sentinel-1A data unavailable (early days of ESA Sentinel-1 data provision)
20150803	Sentinel-1A data unavailable. Link: Sentinel Online News 20150803
20160618 20160619 20160620 20160622 20160623 20160625 20160626	Sentinel-1A data unavailable. Link: Sentinel Online News 20160616, 20160617, 20160620, 20160622, 20160627, 20160628

Table 10 – Dataset files missing for the Arctic Ocean - High Resolution Sea Ice Information L4 (SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_024) product.

Arctic Ocean - High Resolution Sea Ice Information L4 SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024 cmems_obs-si_arc_phy_my_l4_P1D and cmems_obs-si_arc_phy_nrt_l4_P1D	
Date of dataset file [YYYYMMDD]	**Note**
20141015 20141016 20141119	DMI-ASIP L3 product not processed (see Table 9).
20150803	DMI-ASIP L3 product not processed (see Table 9).
20160618 20160619 20160620 20160622 20160623 20160625 20160626	DMI-ASIP L3 product not processed (see Table 9).
20230520 20230521	OSI-458 ICDR* products not processed.

k) Product use and impact

A growing maritime user group accessing wider parts of the Arctic due to sea ice retreat, calls for more effective ways of producing detailed and timely ice information and reliable ice forecasts to the users for maritime safety and planning. To be able to provide these, there is a need for high-resolution ice observations of the Arctic with associated uncertainties for e.g. integration into forecast models.

Manual ice charts from manual multi-sensor satellite data analysis, with synthetic aperture radar (SAR) imagery as primary input data (due to their high-resolution and capability to see through clouds and in polar darkness), has for many years been the preferred method at National Ice Centers around the world for producing sea ice information for maritime safety. Due to their apparent high level of detail (native ice charts come with no quality or uncertainty information) ice chart products have been made available to Model Forecast Centers for assimilation and verification purposes since the days of the MyOcean project (EU project 2009-2014, precursor to the CMEMS). Manual ice charting is a time-consuming method that is increasingly challenged by the increasing volumes of available and free satellite imagery in recent years from Earth Observation programmes such as the EU Copernicus. This challenge will be even more pronounced in the coming years with the planned expansion of the Copernicus program with new SAR and passive microwave satellites such as the Copernicus Imaging Microwave Radiometer (CIMR) and Radar Observation System for Europe in L-band (ROSE-L). Along with a growing maritime user group accessing wider parts of the Arctic due to the retreat and thinning of the Arctic sea ice, this calls for a more effective way of producing detailed and timely ice information to the users. In between the traditional manually produced Ice Charts, covering a subset of the Polar regions, and the daily sea ice products from satellite passive microwave radiometer (PMR) data in coarse resolution e.g from EUMETSAT Ocean and Sea Ice Satellite Applications Facility (OSI SAF) [OSI SAF CDR, 2024], there is a lack of high-resolution sea ice products delivered in near-real time. Ultimately, users want reliable ice forecasts for the coastal areas for e.g. safe navigation and planning. To be able to provide these, there is a need for high-resolution, detailed, consistent and standardized satellite ice observations for the pan-Arctic - and ultimately for both hemispheres - with associated uncertainties for integration in the forecast models.

Recent years have witnessed extraordinary developments in data analytics techniques such as AI, leading to new possibilities in collecting and analyzing the large volumes of data made available by EO satellites e.g. from Copernicus. Service providers in the Copernicus Marine Service (CMS) sea ice TAC have developed algorithms for computation of sea ice information from SAR satellite data, that was integrated into operational processing systems e.g. as the regional datasets of the product "Automatic multi-sensor high resolution ice concentration and type product" available in CMS since 2021 (https://doi.org/10.48670/moi-00122). The products that are subject to this Product User Manual are a further evolution of the SI TAC SAR satellite-based products, developed using state-of-the-art AI methodologies and applicable for a wider part of the Arctic, thus replacing the existing regional datasets.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1 FILE SIZE

Arctic Ocean - High Resolution Sea Ice Information L3 SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023		
DATASET NAME	**NAME OF FILE** *YYYYMMDD	**DIMENSION [GB]**
cmems_obs-si_arc_phy_my_l3_P1D	dmi_asip_seaice_mosaic_arc_l3_*.nc	0.016
cmems_obs-si_arc_phy_nrt_l3_P1D	dmi_asip_seaice_mosaic_arc_l3_*.nc	0.016

Arctic Ocean - High Resolution Sea Ice Information L4 SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024		
DATASET NAME	**NAME OF FILE** *YYYYMMDD	**DIMENSION [GB]**
cmems_obs-si_arc_phy_my_l4_P1D	dmi_asip_seaice_mosaic_arc_l4_*.nc	0.012
cmems_obs-si_arc_phy_nrt_l4_P1D	dmi_asip_seaice_mosaic_arc_l4_*.nc	0.012

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

[QUID, 2024] Quality Information Document for Arctic Ocean - High Resolution Sea Ice Information L3 and L4, SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023, SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024, v1.0, Copernicus Marine Service, 2024, https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SEAICE-QUID-011-023-024.pdf

[011_012 PUM, 2022] Product User Manual for Antarctic Ocean high resolution sea ice product SEAICE_ANT_PHY_AUTO_L3_NRT_011_012, v. 1.0, Copernicus Marine Service, 2022, https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SEAICE-PUM-011-012.pdf

[OSI SAF CDR PUM, 2023] Global Sea Ice Concentration Climate Data Records Product User's Manual, Document version: 3.1, OSI-450-a, OSI-430-a, OSI-458, EUMETSAT Ocean and Sea Ice Satellite Application Facility, April 2023, https://osisaf-hl.met.no/sites/osisaf-hl/files/user_manuals/osisaf_cdop3_ss2_pum_sea-ice-conc-climate-data-record_v3p1.pdf

[OSI SAF NRT PUM, 2023] Product User Manual for Global Sea Ice Concentration Level 2 and Level 3, OSI-410-a, OSI-401-d, OSI-408-a, Document Version: 1.2, EUMETSAT Ocean and Sea Ice Satellite Application Facility, June 2023, https://osisaf-hl.met.no/sites/osisaf-hl/files/user_manuals/osisaf_pum_ice-conc_l2-3_v1p2.pdf

[Wulf, 2024(1)] Wulf, T., Buus-Hinkler, J., Singha, S., Shi, H., and Kreiner, M. B. (2024): Pan-Arctic Sea Ice Concentration from SAR and Passive Microwave, EGUsphere [preprint], September 2024, https://doi.org/10.5194/egusphere-2024-178

[Wulf, 2024(2)] Wulf, T., Buus-Hinkler, J., Singha, S. and Kreiner, M. B. (2024): Operational SAR-Based Sea ICE Concentration Products for Copernicus Marine Service, IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium, July 2024, https://ieeexplore.ieee.org/abstract/document/10641836

6) ANNEX

Arctic Ocean - High Resolution Sea Ice Information L3
SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023
cmems_obs-si_arc_phy_my_l3_P1D and cmems_obs-si_arc_phy_nrt_l3_P1D

```
netcdf file: dmi_asip_seaice_mosaic_arc_l3_20231231.nc {
  dimensions:
    time = 1;
    xc = 15200;
    yc = 22400;
  variables:
    double time(time=1);
      :standard_name = "time";
      :units = "days since 1970-01-01 00:00:00";
      :calendar = "standard";
      :_ChunkSizes = 1U; // uint

    float crs;
      :grid_mapping_name = "polar_stereographic";
      :long_name = "coordinate reference system";
      :false_easting = 0.0; // double
      :false_northing = 0.0; // double
      :proj4_string = "+proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +x_0=0 +y_0=0 +a=6378273
+b=6356889.449 +units=m +no_defs";
      :straight_vertical_longitude_from_pole = -45L; // long
      :latitude_of_projection_origin = 90L; // long
      :standard_parallel = 70L; // long
      :semi_major_axis = 6378273.0; // double
      :semi_minor_axis = 6356889.449; // double
      :spatial_ref = "EPSG:3411";

    double xc(xc=15200);
      :axis = "X";
      :standard_name = "projection_x_coordinate";
      :long_name = "x-coordinate in cartesian system";
      :units = "m";
      :_ChunkSizes = 15200U; // uint

    double yc(yc=22400);
      :axis = "Y";
      :standard_name = "projection_y_coordinate";
      :long_name = "y-coordinate in cartesian system";
      :units = "m";
      :_ChunkSizes = 22400U; // uint

    float lat(yc=22400, xc=15200);
```

```
      :least_significant_digit = 3L; // long
      :long_name = "latitude coordinate";
      :standard_name = "latitude";
      :units = "degrees_north";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :_ChunkSizes = 2240U, 1520U; // uint

    float lon(yc=22400, xc=15200);
      :least_significant_digit = 3L; // long
      :standard_name = "longitude";
      :long_name = "longitude coordinate";
      :units = "degrees_east";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :_ChunkSizes = 2240U, 1520U; // uint

    float acq_time(time=1, yc=22400, xc=15200);
      :_FillValue = 0.0f; // float
      :long_name = "acquisition time (seconds since 1970-01-01 00:00:00)";
      :units = "s";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :_ChunkSizes = 1U, 4480U, 3040U; // uint

    byte sic(time=1, yc=22400, xc=15200);
      :_FillValue = -1B; // byte
      :long_name = "sea ice concentration";
      :standard_name = "sea_ice_area_fraction";
      :units = "%";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :source = "DMI-ASIP";
      :_ChunkSizes = 1U, 7467U, 5067U; // uint

    byte sod(time=1, yc=22400, xc=15200);
      :_FillValue = -1B; // byte
      :long_name = "sea ice stage of development category";
      :units = "1";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :source = "DMI-ASIP";
      :description = "Stage of development category IDs:\n0: New and Young Ice (<30 cm)\n1: Thin First
Year Ice (30 cm - 70 cm)\n2: Thick First Year Ice (>70 cm)\n3: Multi-Year Ice\n4: Glacier Ice";
      :_ChunkSizes = 1U, 7467U, 5067U; // uint

    byte flz(time=1, yc=22400, xc=15200);
      :_FillValue = -1B; // byte
```

```
      :long_name = "sea ice floe size category";
      :units = "1";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :source = "DMI-ASIP";
      :description = "Floe size category IDs:\n0: Small Floes (<100 m)\n1: Medium Floes (100 m - 500 m)\n2: Big Floes (500 m - 2 km)\n3: Vast and Giant Floes (>2 km)";
      :_ChunkSizes = 1U, 7467U, 5067U; // uint

    byte standard_deviation_sic(time=1, yc=22400, xc=15200);
      :_FillValue = -1B; // byte
      :long_name = "standard deviation for the predicted sea ice concentration";
      :units = "%";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :source = "DMI-ASIP";
      :_ChunkSizes = 1U, 7467U, 5067U; // uint

    byte confidence_sod(time=1, yc=22400, xc=15200);
      :_FillValue = -1B; // byte
      :long_name = "confidence level for the predicted stage of development";
      :units = "%";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :source = "DMI-ASIP";
      :_ChunkSizes = 1U, 7467U, 5067U; // uint

    byte confidence_flz(time=1, yc=22400, xc=15200);
      :_FillValue = -1B; // byte
      :long_name = "confidence level for the predicted floe size category";
      :units = "%";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :source = "DMI-ASIP";
      :_ChunkSizes = 1U, 7467U, 5067U; // uint

    ushort status_flag(time=1, yc=22400, xc=15200);
      :flag_descriptions = "no bits set (0): nominal value of the sea ice parameters (sic, sod, flz) retrieved by DMI-ASIP\nbit 1 (1) Land mask\nbit 2 (10): Not in use\nbit 3 (100): Not in use\nbit 4 (1000): Not in use\nbit 5 (10000): Not in use\nbit 6 (100000): Not in use\nbit 7 (1000000): Not in use\nbit 8 (10000000): Not in use\nbit 9 (100000000): Not in use\nbit 10 (1000000000): SOD changed according to SOD-climatology\nbit 11 (10000000000): Maximum sea ice extent climatology screening applied\nbit 12 (100000000000): Not in use\nbit 13 (1000000000000): Not in use\nbit 14 (10000000000000): Not in use\nbit 15 (100000000000000): Not in use\nbit 16 (1000000000000000): Not in use";
      :standard_name = "status_flag";
      :units = "1";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :_ChunkSizes = 1U, 5600U, 3800U; // uint
```

```
  // global attributes:
  :Conventions = "CF-1.4";
  :title = "DMI-ASIP sea ice information";
  :start_date = "2023-12-31 03:22:40Z";
  :stop_date = "2023-12-31 21:20:00Z";
  :product_status = "Operational";
  :abstract = "The sea ice information in this daily mosaic is automatically retrieved from Synthetic
Aperture Radar (SAR) imagery. The retrieval is run at 80 m grid resolution using a Convolutional
Neural Network (CNN) technique. The 80 m grid resolution approximately matches the nominal spatial
resolution of the Copernicus Sentinel-1 imagery in Extra Wide (EW) swath acquisition mode. The CNN
model combines the input SAR imagery with passive microwave data from the AMSR2 sensor onboard the
Japanese GCOM-W satellite.";
  :landmask = "Indication of land: bit 1 set in the 16 bit status_flag variable\nSource:
OpenStreetMap®";
  :activity_type = "Space borne instrument(s)";
  :area = "Arctic";
  :spatial_resolution_latitude = "500.0";
  :spatial_resolution_longitude = "500.0";
  :southernmost_latitude = "40";
  :northernmost_latitude = "90";
  :westernmost_longitude = "-180";
  :easternmost_longitude = "180";
  :institution = "Danish Meteorological Institute (DMI)";
  :institution_reference = "http://www.dmi.dk";
  :contact = "asip-team@dmi.dk";
  :software_version = "ASIP 2.0";
  :history = "Created 2024-09-20 20:18:00 UTC";
}
```

Arctic Ocean - High Resolution Sea Ice Information L4
SEAICE_ARC_PHY_AUTO_L4_MYNRT_011_024
cmems_obs-si_arc_phy_my_l4_P1D and cmems_obs-si_arc_phy_nrt_l4_P1D

```
netcdf file: dmi_asip_seaice_mosaic_arc_l4_20231231.nc {
  dimensions:
    time = 1;
    xc = 7600;
    yc = 11200;
  variables:
    double time(time=1);
      :standard_name = "time";
      :units = "days since 1970-01-01 00:00:00";
      :calendar = "standard";
      :_ChunkSizes = 1U; // uint

    float crs;
      :grid_mapping_name = "polar_stereographic";
      :long_name = "coordinate reference system";
      :false_easting = 0.0; // double
      :false_northing = 0.0; // double
```

```
      :proj4_string = "+proj=stere +lat_0=90 +lat_ts=70 +lon_0=-45 +x_0=0 +y_0=0 +a=6378273
+b=6356889.449 +units=m +no_defs";
      :straight_vertical_longitude_from_pole = -45L; // long
      :latitude_of_projection_origin = 90L; // long
      :standard_parallel = 70L; // long
      :semi_major_axis = 6378273.0; // double
      :semi_minor_axis = 6356889.449; // double
      :spatial_ref = "EPSG:3411";

    double xc(xc=7600);
      :axis = "X";
      :standard_name = "projection_x_coordinate";
      :long_name = "x-coordinate in cartesian system";
      :units = "m";
      :_ChunkSizes = 7600U; // uint

    double yc(yc=11200);
      :axis = "Y";
      :standard_name = "projection_y_coordinate";
      :long_name = "y-coordinate in cartesian system";
      :units = "m";
      :_ChunkSizes = 11200U; // uint

    float lat(yc=11200, xc=7600);
      :least_significant_digit = 3L; // long
      :long_name = "latitude coordinate";
      :standard_name = "latitude";
      :units = "degrees_north";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :_ChunkSizes = 2240U, 1520U; // uint

    float lon(yc=11200, xc=7600);
      :least_significant_digit = 3L; // long
      :standard_name = "longitude";
      :long_name = "longitude coordinate";
      :units = "degrees_east";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :_ChunkSizes = 2240U, 1520U; // uint

    byte sic(time=1, yc=11200, xc=7600);
      :_FillValue = -1B; // byte
      :long_name = "sea ice concentration";
      :standard_name = "sea_ice_area_fraction";
      :units = "%";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
```

```
      :source = "DMI ASIP / OSI SAF EUMETSAT";
      :_ChunkSizes = 1U, 5600U, 3800U; // uint

    byte uncertainty_sic(time=1, yc=11200, xc=7600);
      :_FillValue = -1B; // byte
      :long_name = "standard deviation for the predicted sea ice concentration (DMI-ASIP) / estimate of the total uncertainty for the sea ice concentration (in gaps filled by the OSI SAF algorithm)";
      :units = "%";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :source = "DMI-ASIP / OSI SAF EUMETSAT";
      :_ChunkSizes = 1U, 5600U, 3800U; // uint

    ushort status_flag(time=1, yc=11200, xc=7600);
      :standard_name = "status_flag";
      :flag_descriptions = "no bits set (0): nominal value of the sea ice concentration (sic) retrieved by DMI-ASIP\nbit 1 (1) Land mask\nbit 2 (10): [OSI SAF] Lake mask\nbit 3 (100): [OSI SAF] SIC is set to zero by the open water filter\nbit 4 (1000): [OSI SAF] SIC value is changed for correcting land spill-over effects\nbit 5 (10000): [OSI SAF] Handle with caution, the 2m air temperature is high at this position, and this might be false ice\nbit 6 (100000): [OSI SAF] Value is the result of spatial interpolation\nbit 7 (1000000): [OSI SAF] Value is the result of temporal interpolation\nbit 8 (10000000): [OSI SAF] SIC is set to zero since position is outside maximum sea ice climatology\nbit 9 (100000000): Gapfilled with OSI SAF level-3 sea ice concentration\nbit 10 (1000000000): Not in use\nbit 11 (10000000000): [DMI-ASIP] Maximum sea ice extent climatology screening applied\nbit 12 (100000000000): Not in use\nbit 13 (1000000000000): Not in use\nbit 14 (10000000000000): Not in use\nbit 15 (100000000000000): Not in use\nbit 16 (1000000000000000): Not in use";
      :coordinates = "lon lat";
      :grid_mapping = "crs";
      :units = "1";
      :_ChunkSizes = 1U, 3734U, 2534U; // uint

  // global attributes:
  :Conventions = "CF-1.4";
  :title = "DMI-ASIP / OSI SAF EUMETSAT sea ice information";
  :start_date = "2023-12-31 00:00 Z";
  :stop_date = "2023-12-31 23:59 Z";
  :product_status = "Operational";
  :abstract = "The sea ice information in this daily level-4 mosaic is generated using high resolution sea ice concentration from level-3 DMI-ASIP products -gap-filled with sea ice concentration retrieved by the OSI SAF hybrid algorithm (with status_flag filter applied). The level-3 DMI-ASIP products are automatically retrieved from Synthetic Aperture Radar (SAR) satellite imagery. The retrieval is run at 80 m grid resolution using a Convolutional Neural Network (CNN) technique. The 80 m grid resolution approximately matches the nominal spatial resolution of the Copernicus Sentinel-1 imagery in Extra Wide (EW) swath acquisition mode. The CNN model combines the input SAR imagery with passive microwave data from the AMSR2 sensor onboard the Japanese GCOM-W satellite.";
  :landmask = "Indication of land: bit 1 set in status_flag variable\nSource: OpenStreetMap®";
  :activity_type = "Space borne instrument(s)";
  :area = "Arctic";
  :spatial_resolution_latitude = "1000.0";
  :spatial_resolution_longitude = "1000.0";
  :southernmost_latitude = "40";
  :northernmost_latitude = "90";
  :westernmost_longitude = "-180";
  :easternmost_longitude = "180";
  :institution = "Danish Meteorological Institute (DMI)";
```

```
  :institution_reference = "http://www.dmi.dk";
  :contact = "asip-team@dmi.dk";
  :software_version = "ASIP 2.0  & OSISAF 3.0";
  :history = "Created 2024-09-20 20:23:10 UTC";
}
```